Homework Assignment 1: Slab-Geometry Boltzmann Equation

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1 Introduction & Case's Method

The problem is the monoenergetic, steady-state Boltzmann equation in slab geometry, for an infinite homogeneous isotropic medium with an isotropic plane source at $x = 0$:

$$\mu\frac{\partial\psi(x,\mu)}{\partial x} + \psi(x,\mu) = \frac{c}{2}\int_{-1}^{1}\psi(x,\mu')d\mu' + \frac{1}{4\pi}\delta(x) \tag{1}$$

with $c \in [0, 1)$ the scattering ratio and μ the direction cosine. Case (1953) looks for separable solutions $\psi_\nu(x,\mu) = e^{-x/\nu}\phi_\nu(\mu)$, which gives

$$(\nu-\mu)\phi_\nu(\mu) = \frac{c\nu}{2}\int_{-1}^{1}\phi_\nu(\mu')d\mu' \quad \xrightarrow{\int\phi_\nu d\mu=1} \quad (\nu-\mu)\phi_\nu(\mu) = \frac{c\nu}{2} \tag{2}$$

and splits into two spectra:

1. **Discrete ($\nu \notin [-1, 1]$).** Division by $\nu-\mu$ is legal, so $\phi_0^{\pm}(\mu) = c\nu_0/[2(\nu_0\mp\mu)]$, and normalization gives the characteristic equation

$$1 = c\nu_0 \operatorname{arctanh}(1/\nu_0) \tag{3}$$

   with exactly one pair of real roots $\pm\nu_0$, $\nu_0 > 1$, for $c \in (0, 1)$.

2. **Continuous ($\nu \in [-1, 1]$).** Here $\nu - \mu$ vanishes at $\mu = \nu$, so the solution exists only as a distribution,

$$\phi_\nu(\mu) = \frac{c\nu}{2}\mathcal{P}\frac{1}{\nu-\mu} + \lambda(\nu)\delta(\mu-\nu), \qquad \lambda(\nu) = 1 - c\nu\operatorname{arctanh}(\nu) \tag{4}$$

   with $\mathcal{P}$ the Cauchy principal value and λ fixed by the same normalization.

The singular objects live in angle only. The scalar flux $\phi(x) = 2\pi\int_{-1}^{1}\psi\, d\mu$ integrates them out, since $\int_{-1}^{1}\phi_\nu(\mu)d\mu = 1$ by construction, leaving the ordinary integral

$$\phi_{\mathrm{tr}}(x) = \frac{1}{2}\int_0^1\frac{e^{-|x|/\nu}}{N_\nu}d\nu, \qquad N_\nu = \nu\left[\lambda^2(\nu) + \frac{\pi^2c^2\nu^2}{4}\right] \tag{5}$$

whose integrand is smooth on $(0, 1)$: it vanishes at $\nu \to 0$ through $e^{-|x|/\nu}$ and at $\nu \to 1$ through $\lambda \to -\infty$, so the quadrature is well behaved.

2 Question 1: Exact Scalar Flux Analysis

The exact flux is the sum of the two contributions,

$$\phi(x) = \phi_{\rm as}(x) + \phi_{\rm tr}(x), \qquad \phi_{\rm as}(x) = \frac{e^{-|x|/\nu_0}}{2N_0^+}, \qquad N_0^+ = \frac{c}{2}\nu_0^3\left[\frac{c}{\nu_0^2-1} - \frac{1}{\nu_0^2}\right] \tag{6}$$

ν_0 is computed two ways: by bisection on $\operatorname{arctanh}(y)/y = 1/c$ with $y = 1/\nu_0 \in (0,1)$, and from the analytic fit

$$\nu_0 \approx \frac{1}{\sqrt{1-c^{p(c)}}}, \qquad p(c) = 2.47412 + \frac{0.00363081}{c^2} - 0.0352458c + \frac{0.557498}{c} \tag{7}$$

which agree to better than 5×10^{-3} % over $c = 0.5 \ldots 0.95$. All figures are produced with both.

2.1 Question 1a: Asymptotic and Transient Components

[Insert plots of asymptotic and transient components for each c, and the relative contributions $\phi_{\rm tr}(x)/\phi(x)$ and $\phi_{\rm as}(x)/\phi(x)$ as a function of x. Add physical discussion here.]

2.2 Question 1b: Classical Diffusion Solution

In mean free paths ($\Sigma_t = 1$) the diffusion equation for the same source is

$$-D\phi''(x) + (1-c)\phi(x) = \delta(x) \tag{8}$$

and with the P_1 value $D = 1/3$ its Green's function is

$$\phi_{\rm class}(x) = \frac{\sqrt{3}}{2\sqrt{1-c}} e^{-\sqrt{3(1-c)}|x|} \tag{9}$$

[Insert comparison plots with the classical diffusion solution. Discuss the differences, including the behavior near the source ($x \to 0$) and the asymptotic behavior ($x \to \infty$).]

2.3 Question 1c: Asymptotic Diffusion Solution

Matching the diffusion length to the exact eigenvalue, $L = \nu_0$, fixes $D_{\rm as} = (1-c)\nu_0^2$ and gives

$$\phi_{\rm as,diff}(x) = \frac{1}{2(1-c)\nu_0} e^{-|x|/\nu_0} \tag{10}$$

which is exactly Case's asymptotic mode: this approximation reproduces $\phi_{\rm as}$ by construction and errs only by the transient part.

[Insert comparison plots with the asymptotic diffusion solution. Discuss how the asymptotic diffusion solution compares to both the classical diffusion solution and the exact transport asymptotic component $\phi_{\rm as}(x)$.]

2.4 Question 1d: Relative Error of Diffusion Approximations

$$\text{Relative Error (\%)} = \frac{|\phi_{\rm approx}(x) - \phi_{\rm exact}(x)|}{\phi_{\rm exact}(x)} \times 100\% \tag{11}$$

[Insert relative error plots as a function of x. Add physical discussion comparing the accuracy of the classical and asymptotic diffusion approximations across the different values of c. Analyze why the error behaves as it does near the source and far from it.]

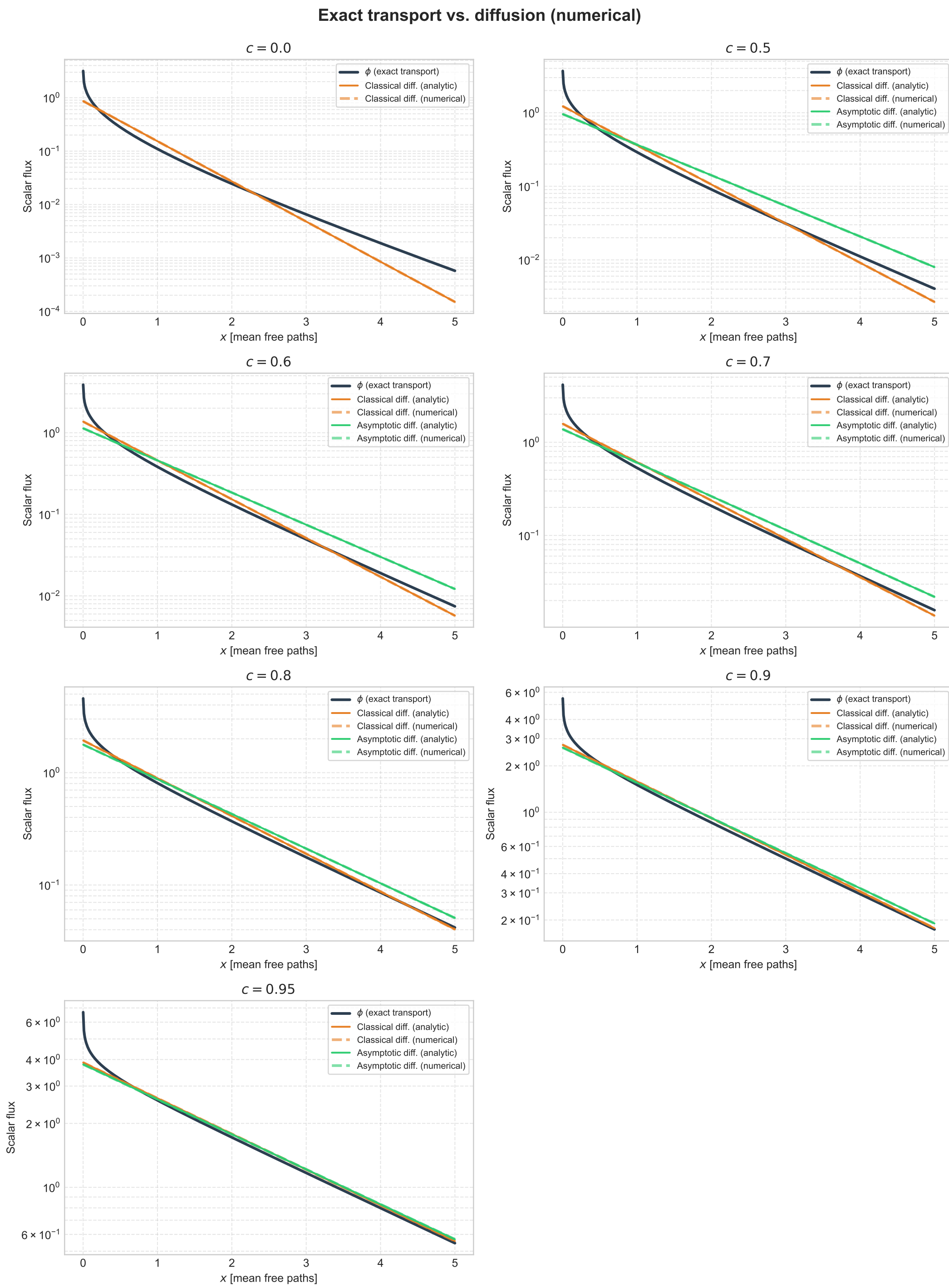

Figure 1: Exact scalar flux against the classical and asymptotic diffusion approximations, for $c = 0, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95$. Each approximation appears both as its closed form and as the numerical solution of Question 2; the two agree to 2×10^{-8} % and are indistinguishable here.

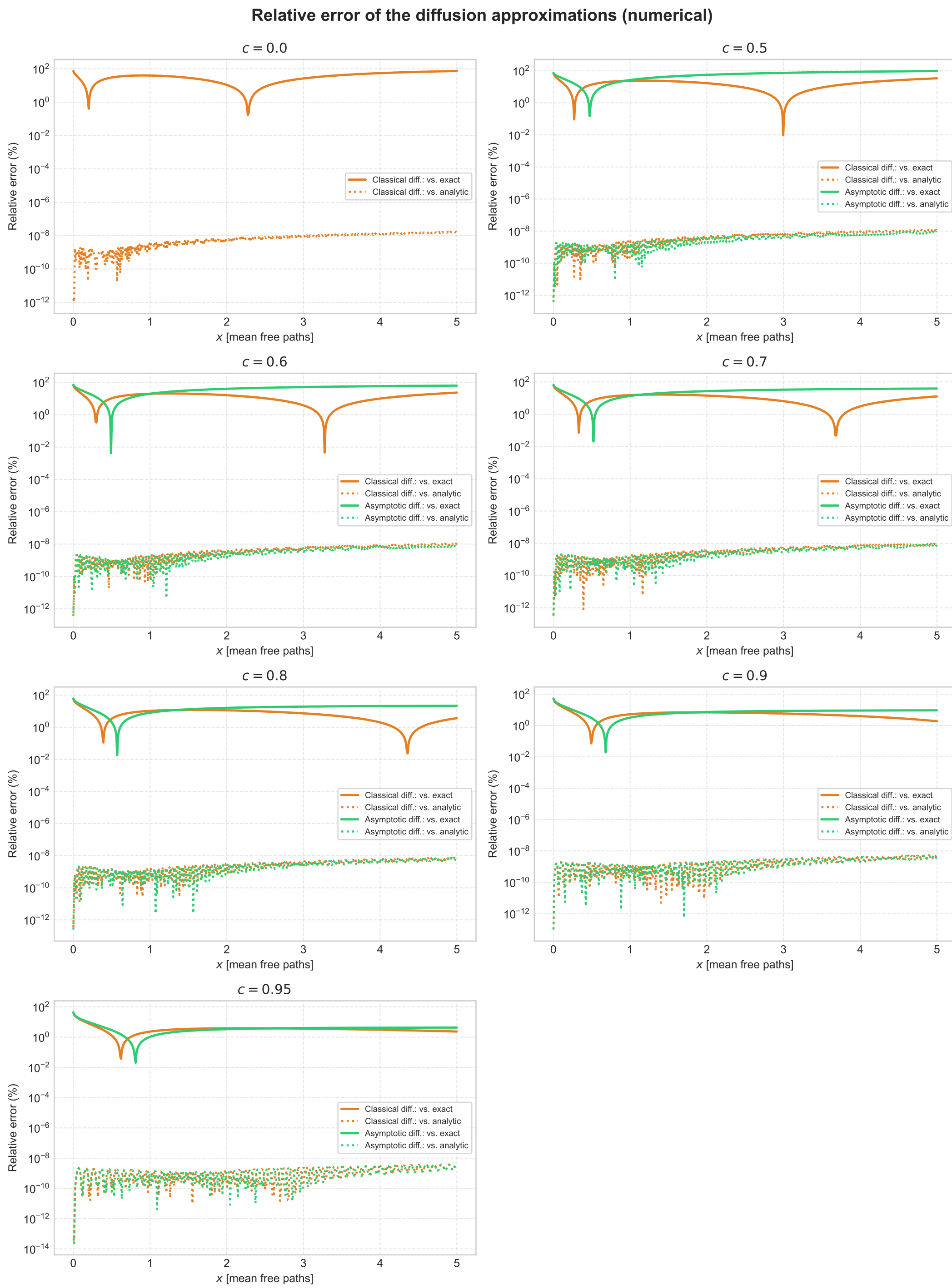

Figure 2: Two errors per approximation: against the *exact transport* solution (solid), which is the modelling error asked for in 1d, and against the *analytic diffusion* solution (dotted), which is the solver error asked for in Question 2. They differ by about ten orders of magnitude, so the solid curves may be read as properties of the approximation rather than of the code.

3 Question 2: Numerical Code for the Diffusion Equation

The medium is symmetric, so the problem is solved on the half-domain $x \in [0, a]$, where it is source-free:

$$-D\phi''(x) + (1-c)\phi(x) = 0 \quad \Longrightarrow \quad \frac{d^4}{dx}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} y_2 \\ \frac{1-c}{D} y_1 \end{pmatrix}, \qquad y_1 = \phi,\ y_2 = \phi' \tag{12}$$

The delta source. It is never discretised. Integrating across the origin gives $-D[\phi'(0^+)-\phi'(0^-)] = 1$, and evenness of the infinite-medium solution gives

$$\phi'(0^+) = -\frac{1}{2D}, \qquad \text{i.e.} \qquad J(0^+) = -D\phi'(0^+) = \frac{1}{2} \tag{13}$$

half the source streaming each way. That is how a delta source should be modelled: symmetry converts it into a prescribed current and the singularity never enters the scheme.

The outer boundary. $x = a$ is a truncation, not a surface. Zero flux there would force a 100 % error at $x = a$ that no tolerance removes, so the radiation condition

$$\phi'(a) = -\kappa\phi(a), \qquad \kappa = \sqrt{\Sigma_a/D} \tag{14}$$

is used instead: $e^{-\kappa x}$ satisfies it identically, the growing mode is annihilated, and the truncation is exact for any a. We take $a = 10/\kappa$.

One integration. The equation is linear and homogeneous away from the source, so the solution is proportional to its starting amplitude and no root-finding is needed. We integrate once from a to 0 — the direction in which the solution grows, which is the stable one — from $[y_1, y_2]^T = [1, -\kappa]^T$, then rescale by $[-1/(2D)]/y_2(0)$ to enforce the source condition. Both approximations use this one solver, since

$$\Sigma_a = 1 - c, \qquad D_{\text{classical}} = \frac{1}{3}, \qquad D_{\text{asymptotic}} = (1 - c)\,\nu_0^2 \tag{15}$$

Verification. The solver reproduces the closed form to 2×10^{-8} % for both approximations at $c = 0.5, 0.7, 0.9$ (Figures 1–4), and conserves neutrons to $\Sigma_a \int \phi\, dx = 0.99998807$, the deficit being the physical tail beyond the truncation. A shooting method has no mesh to refine, so convergence is shown against the integrator tolerance: the relative L_2 error falls from 9×10^{-6} at `rtol` $= 10^{-4}$ to 1.3×10^{-12} at 10^{-11}, with unit slope (Figure 5).

4 Question 3: Critical Half-Thickness and Critical Radius

Parts 3(a)–3(c) are covered: the exact-transport relations, and classical diffusion with the Marshak and Mark conditions. Parts 3(d)–3(e) are not.

4.1 The Eigenvalue for $c > 1$

The discrete eigenvalue leaves the real axis. With $\nu_0 = i|\nu_0|$, $k_0 = 1/|\nu_0|$ and $\operatorname{arctanh}(ik) = i\arctan(k)$, the characteristic equation becomes real again,

$$c\arctan(k_0) = k_0 \tag{16}$$

the form tabulated by Case, de Hoffmann & Placzek (Table 8, Part II), which the root reproduces to 5×10^{-6}. The fit of Question 1 continues onto this branch by itself, its radicand simply changing sign:

$$|\nu_0(c)| = \frac{1}{\sqrt{c^{p(c)} - 1}} \tag{17}$$

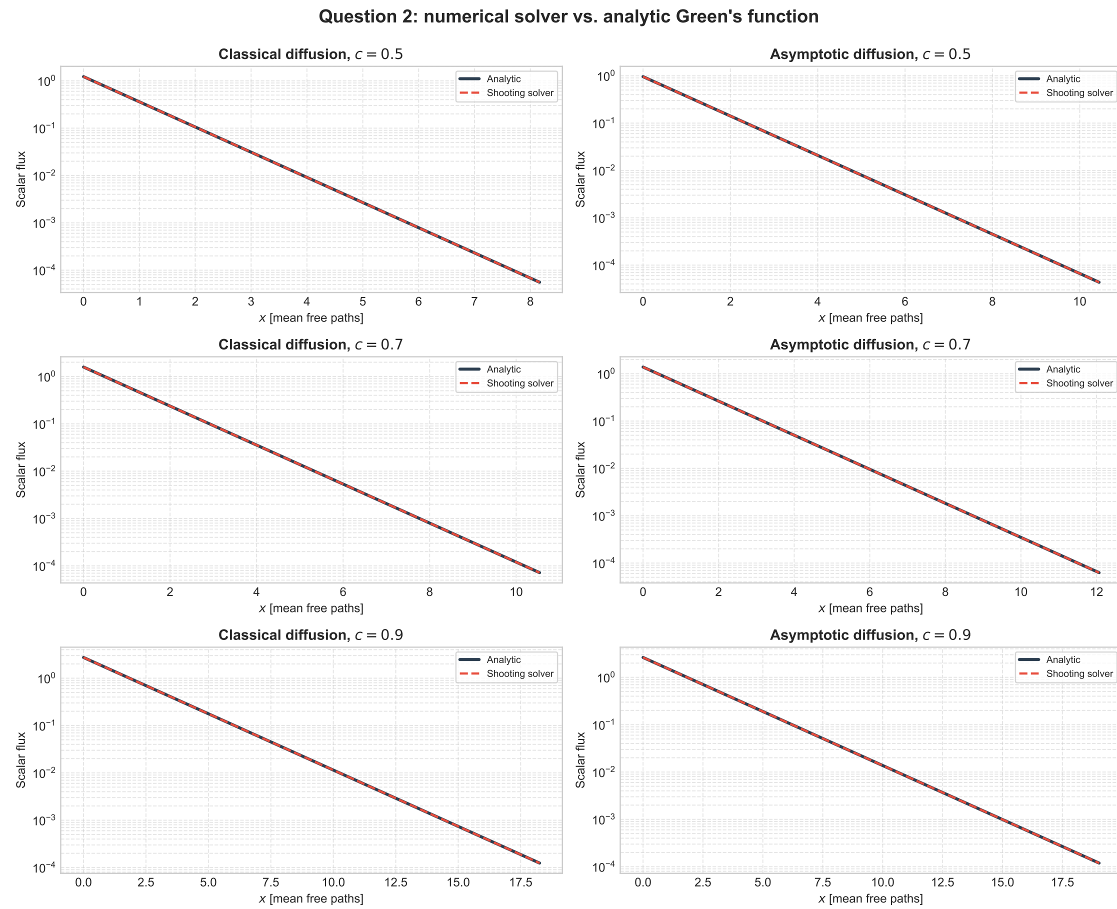

Figure 3: Numerical solution against the analytic diffusion Green's function, both approximations at $c = 0.5, 0.7, 0.9$.

accurate to $10^{-4}\,\%$ at $c = 1.02$ and $0.10\,\%$ at $c = 2$. An imaginary ν_0 makes the asymptotic flux trigonometric, which is what allows a finite critical system:

$$\phi_{\rm as}(x) = A\cos\left(\frac{\Sigma_t x}{|\nu_0|}\right), \qquad \phi_{\rm as}(r) = A\frac{\sin(\Sigma_t r/|\nu_0|)}{\Sigma_t r} \tag{18}$$

the sine being excluded in the slab by symmetry, the cosine in the sphere by regularity.

4.2 The Criticality Relations

The asymptotic flux extrapolates to zero a distance $z_0(c)$ outside the surface. Placing the quarter-wave zero of the cosine and the first zero of the sine there gives, in mean free paths,

$$\frac{a}{2} = \frac{\pi}{2}\,|\nu_0(c)| - z_0(c), \qquad \Sigma_t R_c = \pi\,|\nu_0(c)| - z_0(c) \tag{19}$$

No diffusion coefficient enters, so these are exact transport in that sense; they are still asymptotic, in that the transient flux near the surface is neglected (and the sphere uses the plane Milne z_0, without the Winslow curvature correction). The extrapolation distance comes from the expansion about $c = 1$,

$$z_0(c) = 0.710446\left[\frac{1}{c} + q\,\frac{(1-c)^2}{c} + \mathcal{O}\left(\frac{(1-c)^3}{c}\right)\right] \tag{20}$$

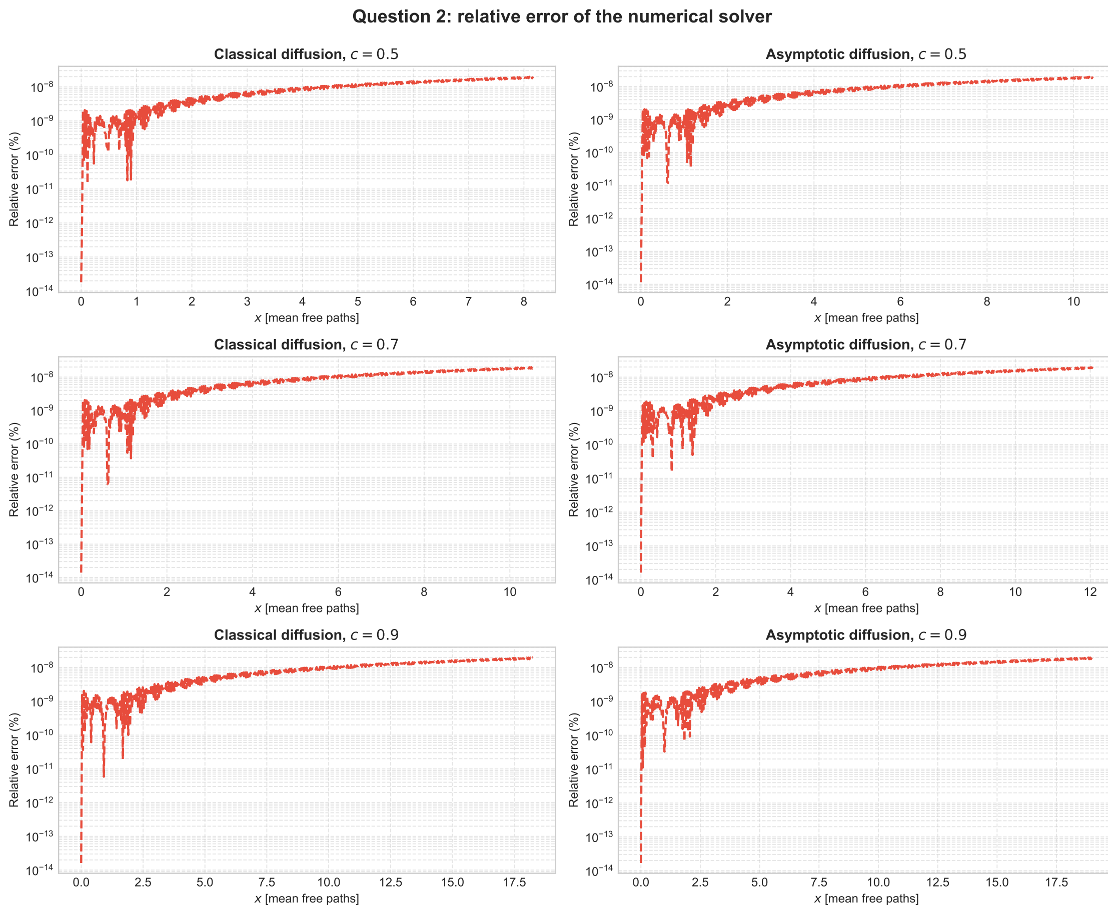

Figure 4: Relative error across the domain. With the radiation condition it stays flat instead of diverging at the outer edge.

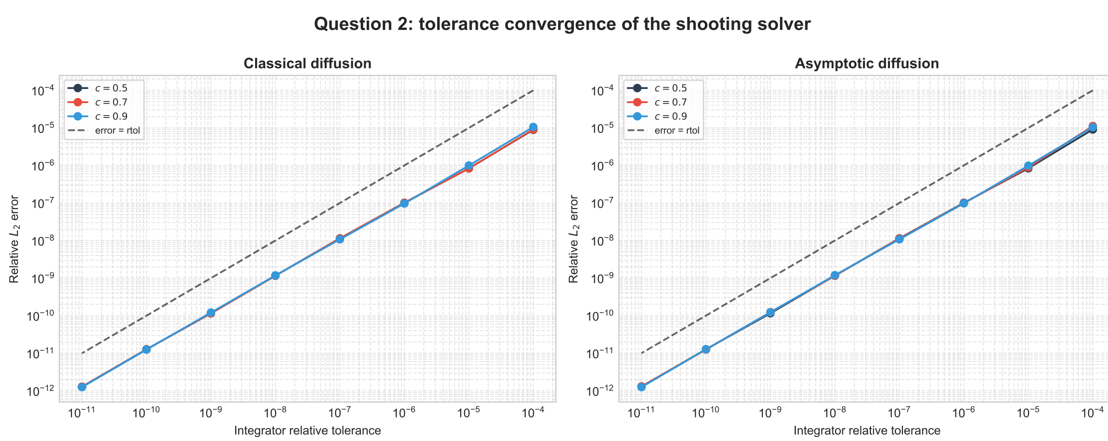

Figure 5: Tolerance convergence of the shooting solver.

whose leading term is exact in the sense that $0.710446/c = 0.710446\,\nu_0 \operatorname{arctanh}(1/\nu_0)$, with 0.710446 Hopf's constant.

4.3 The Sign of the Quadratic Correction

The notes print $q = -0.0199$, but Case's Table 23 shows $c\,z_0(c)$ rising away from 0.710446 on *both* sides of $c = 1$ (Figure 7), which a negative q cannot reproduce; with $q = +0.0199$ the fit matches the table to its four printed digits at $c = 0.9$ and $c = 1.1$. The printed minus sign therefore looks like a transcription error. Over $1.02 < c < 2$ the two signs differ by at most 2.6 % in z_0 and 3.2 % in $a/2$, and by less than 0.07 % in $a/2$ below $c = 1.2$. Results are quoted with the printed sign; both are plotted.

4.4 Results

c	$\lvert\nu_0(c)\rvert$	$z_0(c)$	$a/2$	$\Sigma_t R_c$	error in $a/2$
1.02	4.050190	0.696510	5.665513	12.027537	0.0001 %
1.05	2.531781	0.676582	3.300331	7.277244	0.0025 %
1.10	1.756653	0.645731	2.113612	4.872955	0.0128 %
1.20	1.198272	0.591567	1.290674	3.172916	0.0667 %
1.30	0.946021	0.545518	0.940488	2.426495	0.1777 %
1.40	0.793813	0.505846	0.741072	1.987990	0.3592 %
1.50	0.689210	0.471274	0.611333	1.693941	0.6116 %
1.60	0.611746	0.440848	0.520080	1.481008	0.9535 %
1.70	0.551520	0.413834	0.452491	1.318817	1.3672 %
1.80	0.503065	0.389665	0.400547	1.190759	1.8831 %
1.90	0.463074	0.367892	0.359503	1.086897	2.4781 %
2.00	0.429411	0.348154	0.326364	1.000881	3.1859 %

Table 1: Critical dimensions in mean free paths, from the approximate $|\nu_0(c)|$ and the printed z_0 fit ($q = -0.0199$). The last column is the departure of $a/2$ from a reference using the transcendental root and Case's Table 23.

Both dimensions fall steeply with c, from $a/2 = 5.67$, $\Sigma_t R_c = 12.03$ at $c = 1.02$ to 0.33 and 1.00 at $c = 2$: a barely multiplying system must be many mean free paths across, because $|\nu_0| \to \infty$ as $c \to 1^+$.

The error of the approximate route is dominated by z_0, not by $|\nu_0|$ — the eigenvalue fit stays under 0.11 %, while the two-term z_0 expansion reaches 2.6 % by $c = 2$, where $(1-c)^3$ is no smaller than $(1-c)^2$. Since z_0 is subtracted while $|\nu_0|$ is multiplied by π or $\pi/2$, the same absolute z_0 error hurts $a/2$ about twice as much as R_c, and hurts both more as the system shrinks: under 0.1 % up to $c \approx 1.25$, a few percent by $c = 2$ (Figure 6).

4.5 Parts 3(b) and 3(c): Marshak and Mark Conditions

A multiplying medium has $\Sigma_a = 1 - c < 0$, so classical diffusion gives a Helmholtz equation

$$-D\nabla^2\phi + (1-c)\phi = 0 \quad\Longrightarrow\quad \nabla^2\phi + B^2\phi = 0, \qquad B^2 = \frac{c-1}{D} = 3(c-1) \tag{21}$$

whose solutions are the *same* shapes $\cos(Bx)$ and $\sin(Br)/r$. Diffusion therefore leaves (19) unchanged in form and only replaces the two inputs:

$$|\nu_0(c)| \;\longrightarrow\; \frac{1}{B} = \frac{1}{\sqrt{3(c-1)}}, \qquad z_0(c) \;\longrightarrow\; \begin{cases} 2/3 & \text{Marshak} \\ 1/\sqrt{3} & \text{Mark} \end{cases} \tag{22}$$

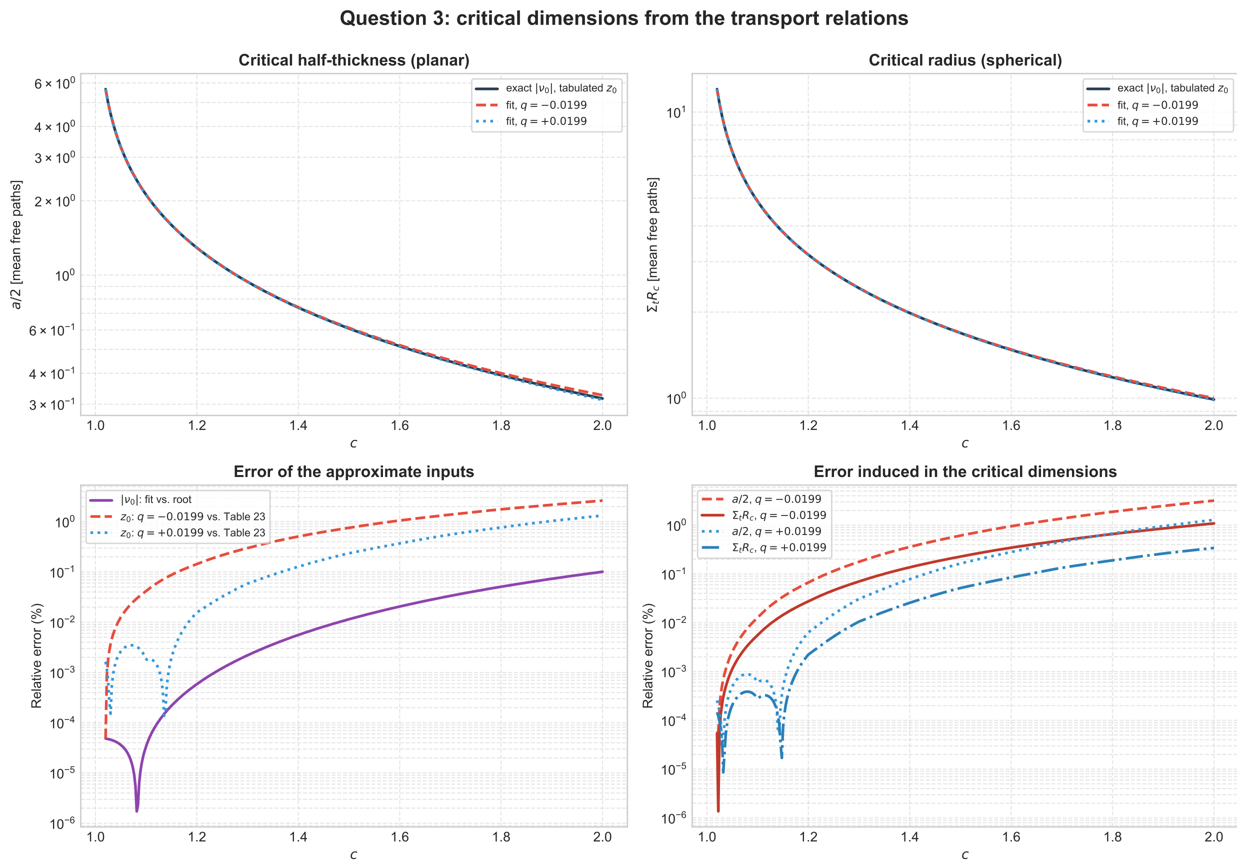

Figure 6: Critical half-thickness and radius for $1.02 < c < 2$ (top), and the errors of the approximate inputs and of the resulting dimensions (bottom).

Marshak sets the incoming partial current to zero, $\phi/4 + D\phi'/2 = 0$, so $z_0 = \ell_0 = 2D = 2/3$; Mark sets the incoming angular flux to zero along $\mu = 1/\sqrt{3}$, so $z_0 = \ell_0 = \sqrt{3}D \approx 0.5774$. Neither input carries the true c-dependence: $1/B$ is only the $c \to 1^+$ limit of $|\nu_0|$, and both z_0 are constants while the transport $z_0(c) \approx 0.7104/c$ halves across the range.

4.6 Comparison

All three agree as $c \to 1^+$, where the system is many mean free paths across and the boundary layer is a small correction, and separate as it shrinks (Figure 8). The two diffusion errors act in opposite directions:

- $1/B$ *over*-estimates the relaxation length — by 0.8 % at $c = 1.02$ and 34 % at $c = 2$ — inflating both dimensions;
- the constant z_0 over-estimates the extrapolation distance as c grows and, being subtracted, deflates both.

The relaxation length is weighted by π in the sphere but only $\pi/2$ in the slab, while z_0 enters both identically. So the spherical radius is over-estimated throughout, by +15 % (Marshak) and +24 % (Mark) at $c = 2$, whereas Marshak's $a/2$ is non-monotonic: +5.5 % at $c \approx 1.23$, zero near $c = 1.52$,

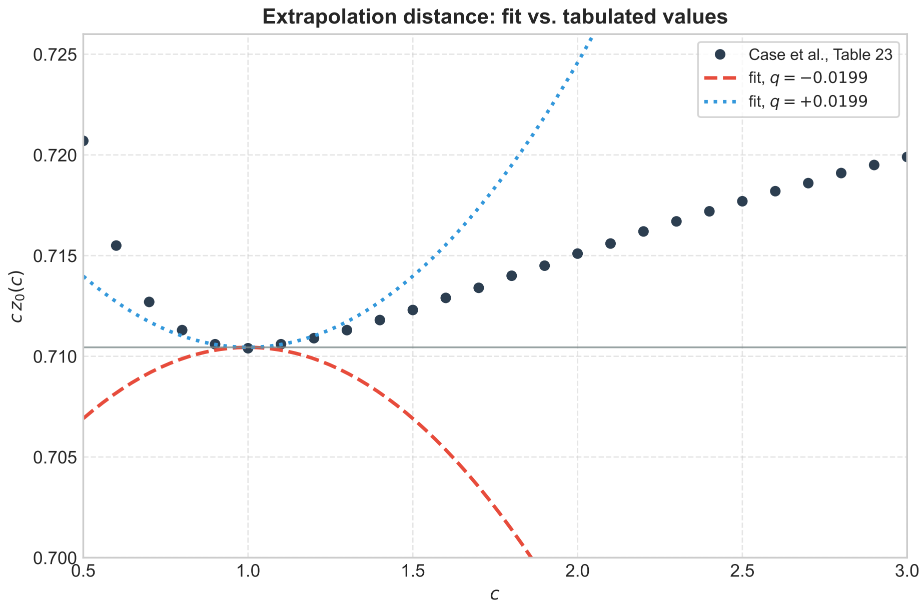

Figure 7: The tabulated product $c\,z_0(c)$ against the fit with either sign of q. The tabulated values rise on both sides of $c = 1$, which fixes the sign as positive.

	$a/2$			$\Sigma_t R_c$		
c	transport	Marshak	Mark	transport	Marshak	Mark
1.02	5.665513	5.746082	5.835399	12.027537	12.158832	12.248148
1.05	3.300331	3.389112	3.478428	7.277244	7.444891	7.534207
1.10	2.113612	2.201202	2.290518	4.872955	5.069071	5.158387
1.20	1.290674	1.361223	1.450539	3.172916	3.389112	3.478428
1.30	0.940488	0.989098	1.078414	2.426495	2.644863	2.734179
1.40	0.741072	0.767268	0.856584	1.987990	2.201202	2.290518
1.50	0.611333	0.615883	0.705200	1.693941	1.898433	1.987749
1.60	0.520080	0.504136	0.593452	1.481008	1.674938	1.764255
1.70	0.452491	0.417286	0.506602	1.318817	1.501238	1.590555
1.80	0.400547	0.347278	0.436594	1.190759	1.361223	1.450539
1.90	0.359503	0.289290	0.378606	1.086897	1.245246	1.334562
2.00	0.326364	0.240233	0.329549	1.000881	1.147133	1.236449

Table 2: Critical dimensions in mean free paths from the three treatments: exact transport (3a), Marshak diffusion (3b) and Mark diffusion (3c).

$-26\,\%$ at $c = 2$. Mark's slab result ends at only $+1\,\%$ at $c = 2$, which is cancellation rather than accuracy.

That reading is confirmed by dropping the shortcut $z_0 = \ell_0$ and imposing $\phi + \ell_0\phi' = 0$ on the flux shape itself,

$$B\,\frac{a}{2} = \arctan\!\left(\frac{1}{B\ell_0}\right), \qquad u\cot u = 1 - \frac{u}{B\ell_0}, \quad u = BR \in (0, \pi) \tag{23}$$

against $Ba/2 = \pi/2 - B\ell_0$ and $u = \pi - B\ell_0$ for the extrapolated form. Since $\arctan(1/y) =$

$\pi/2 - y + \mathcal{O}(y^3)$ the two agree to first order and coincide as $c \to 1$ (0.1 % in the slab at $c = 1.02$), but part company as the system shrinks: at $c = 2$ the Marshak slab gives $a/2 = 0.412$ applied against 0.240 extrapolated. The applied-condition results lie *above* the transport answer in every case.

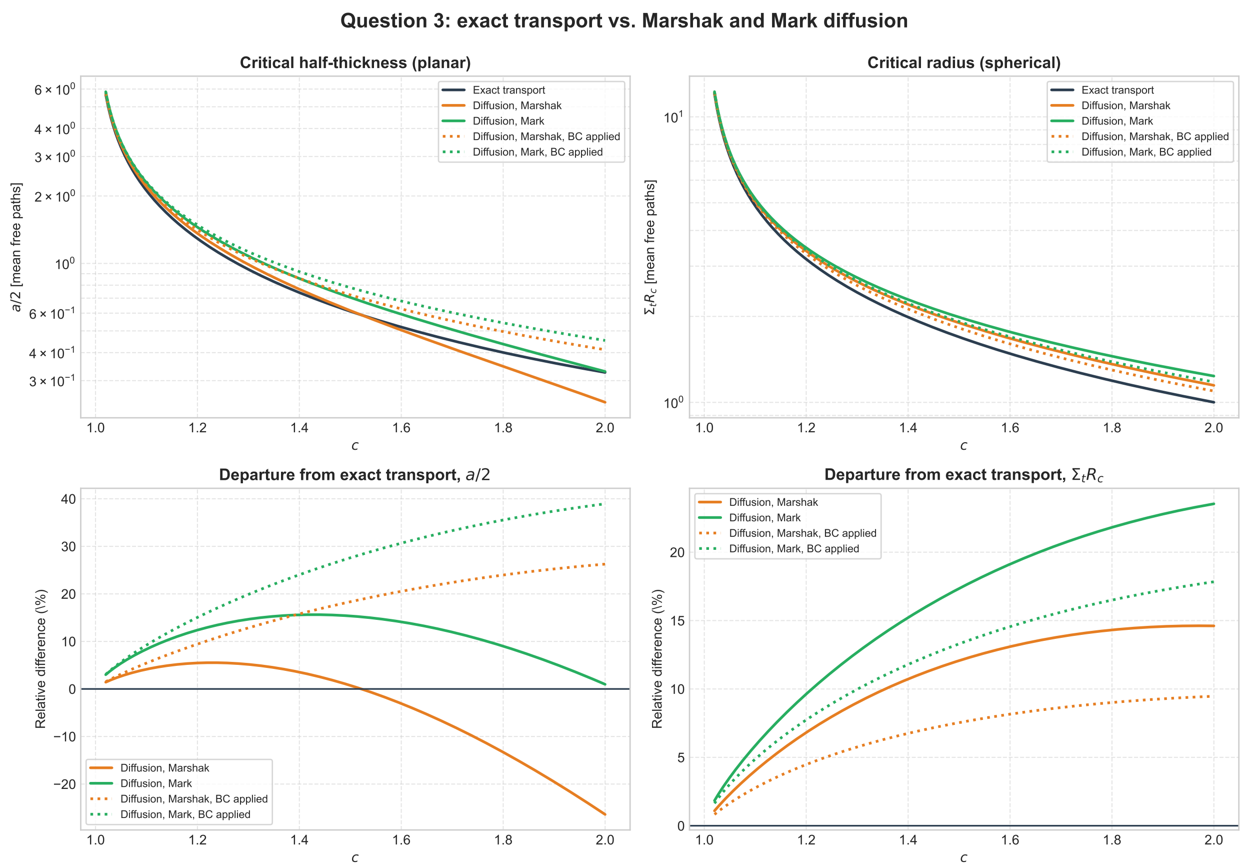

Figure 8: Exact transport against Marshak and Mark diffusion (top), and the signed departure of each from transport (bottom). Dotted curves impose the boundary condition on the flux shape instead of using an extrapolated zero.

5 Question 4: Numerical Diffusion Code in Spherical Coordinates

The code solves

$$-D\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right) + \Sigma_a\phi(r) = \frac{1}{k}\nu\Sigma_f\phi(r) \tag{24}$$

and finds the radius at which $k = 1$. Dividing the fission source by k makes a solution exist at every radius, so criticality is a root search on R rather than an eigenvalue search. Both approximations use the same solver, differing in two constants:

$$\text{classical:} \quad D = \frac{1}{3\Sigma_t}, \quad z_0 = 2D, \qquad \text{asymptotic:} \quad D = \frac{(c-1)|\nu_0(c)|^2}{\Sigma_t}, \quad z_0 = z_0(c) \tag{25}$$

The asymptotic D is the continuation of Question 2's $(1-c)\nu_0^2$ above $c = 1$: both ν_0^2 and $1-c$ change sign, so D stays positive. Its purpose is to give the diffusion equation the true transport relaxation length $|\nu_0|/\Sigma_t$ instead of $1/\sqrt{3(c-1)}$.

5.1 Discretisation

Finite volume on N uniform cell-centred shells. Integrating (24) over cell j turns the divergence into a difference of surface currents,

$$A_{j+1/2}J_{j+1/2} - A_{j-1/2}J_{j-1/2} + \Sigma_a V_j \phi_j = S_j V_j, \qquad J_{j+1/2} = -D\,\frac{\phi_{j+1} - \phi_j}{h} \tag{26}$$

with $A = 4\pi r^2$ and $V_j = \frac{4\pi}{3}(r^3_{j+1/2} - r^3_{j-1/2})$. The balance form is what removes the $1/r^2$ singularity: the innermost face has $A_{1/2} = 0$, so $\phi'(0) = 0$ is imposed exactly with no special case. The matrix is tridiagonal and solved in $O(N)$. Appendix A works the mesh, the index map and the assembly through in full.

Both outer treatments are the single condition $\phi + \ell_0\phi' = 0$; eliminating the surface flux gives one leakage coefficient

$$J_s = \frac{D\,\phi_{N-1}}{\ell_0 + h/2} \tag{27}$$

covering the extrapolated zero ($\ell_0 = 0$ on a mesh run to $R + z_0$) and the Robin condition ($\ell_0 = z_0$ at R) without a branch. The extrapolated form is used, since it is what the Question 3 relations assume, making the comparison a test of the numerics alone.

5.2 The Iterative Algorithm

k comes from the source iteration of Bell & Glasstone, pp. 189–192: start from a flat flux and $k = 1$; form $S = \frac{1}{k}\nu\Sigma_f\phi$; solve the removal problem $(-D\nabla^2 + \Sigma_a)\phi^{(n+1)} = S$; update

$$k^{(n+1)} = k^{(n)}\,\frac{\int \nu\Sigma_f\phi^{(n+1)}\,dV}{\int \nu\Sigma_f\phi^{(n)}\,dV} \tag{28}$$

and repeat until $|k^{(n+1)} - k^{(n)}| < 10^{-10}k^{(n+1)}$. The flux is renormalised each sweep: its amplitude grows by roughly k per sweep, and over several hundred sweeps the convergence test would otherwise lose its significant digits. This is physically irrelevant, since the ratio is homogeneous in ϕ.

Convergence is at the dominance ratio k_1/k_0, which tends to unity for a large weakly multiplying sphere (≈ 0.94 at $c = 1.02$) and is derived in §5.4. The critical radius then follows from Brent bisection on $k(R) - 1$, bracketed within $\pm 5\%$ of the analytic radius so that k is never evaluated far from criticality.

5.3 Comparison with the Analytical Result

At $k = 1$ equation (24) is Helmholtz with $B^2 = (\nu\Sigma_f - \Sigma_a)/D = \Sigma_t(c-1)/D$, and its regular solution $\sin(Br)/r$ vanishes at the extrapolated radius when

$$R_c = \frac{\pi}{B} - z_0 \tag{29}$$

Substituting the two parameter sets recovers the Question 3 relations exactly: $\Sigma_t R_c = \pi/\sqrt{3(c-1)} - 2/3$ classically, and $\Sigma_t R_c = \pi|\nu_0| - z_0(c)$ asymptotically, the latter being part 3(a). Note that R_c depends on the cross sections only through c, since Σ_a and $\nu\Sigma_f$ enter only as their difference.

The residual is discretisation error alone, and the scheme is second order: at $c = 1.5$ the relative error of R_c falls from 5.28×10^{-4} at $N = 25$ to 5.16×10^{-7} at $N = 800$, a factor of exactly 4.00 per doubling (Figure 10). Its sign is consistently negative — the discrete operator is slightly too

Table 3: Critical radius in mean free paths from the $k=1$ search ($N=400$) against (29).

	Classical		Asymptotic		
c	numerical	analytic	numerical	analytic	difference (classical)
1.02	12.15881	12.15883	12.02752	12.02754	-1.6×10^{-4} %
1.10	5.06906	5.06907	4.87295	4.87295	-1.7×10^{-4} %
1.20	3.38911	3.38911	3.17289	3.17289	-1.8×10^{-4} %
1.50	1.89843	1.89843	1.69369	1.69369	-2.1×10^{-4} %
1.80	1.36122	1.36122	1.18995	1.18996	-2.3×10^{-4} %
2.00	1.14713	1.14713	0.99952	0.99952	-2.4×10^{-4} %

leaky on a coarse mesh, so it reaches $k=1$ at a marginally smaller radius. Two further checks: k at the returned radius is 1 to ten digits, and the converged flux matches $\sin(Br)/Br$ to 3.2×10^{-6} (Figure 11).

The two approximations differ from each other far more than either differs from its own analytic solution, and the gap is entirely in the relaxation length: classical diffusion gives the larger radius throughout, by 1.1 % at $c=1.02$ rising to 14.8 % at $c=2$.

5.4 Verification

Three further checks, independent of the comparison in Table 3.

Convergence rate of the iteration. The error of source iteration falls by the dominance ratio k_1/k_0 per sweep. With the spherical modes $B_n = n\pi/r_{\text{out}}$ this is $(\Sigma_a + DB_1^2)/(\Sigma_a + DB_2^2)$, which at a *critical* radius collapses. Criticality means $\Sigma_a + DB_1^2 = \nu\Sigma_f$, and $B_2 = 2B_1$, so

$$\frac{k_1}{k_0} = \frac{\nu\Sigma_f}{4\nu\Sigma_f - 3\Sigma_a} = \frac{c}{4c-3} \tag{30}$$

The diffusion coefficient has cancelled: *the convergence rate at criticality depends on c alone*. This is borne out exactly — the measured ratio agrees with (30) to six digits, and the two approximations take an identical number of sweeps at every c despite their different critical radii.

Table 4: Dominance ratio and sweep count to $|\Delta k| < 10^{-10}k$ at the critical radius. The sweep counts are identical for the classical and asymptotic approximations, as (30) requires.

c	$c/(4c-3)$	measured ratio	sweeps
1.02	0.944444	0.944444	280
1.05	0.875000	0.875000	133
1.10	0.785714	0.785714	79
1.20	0.666667	0.666667	50
1.50	0.500000	0.500000	32
2.00	0.400000	0.400000	25

Neutron balance. For the converged critical flux, production must equal absorption plus leakage. The residual $|P-A-L|/P$ lies between 6×10^{-16} and 1.3×10^{-13} over the whole range, for both approximations. The informative term is the leakage: production and absorption are the same volume integral times a constant, so their ratio $\nu\Sigma_f/(k\Sigma_a)$ holds identically whatever the flux. Leakage, by contrast, is computed from the boundary formula using only ϕ_{N-1}, and nothing forces it to close the balance unless the discrete operator conserves neutrons cell by cell and the boundary coefficient is exactly right. At $c = 1.5$ classical, $P = 32.2340$, $A = 21.4893$, $L = 10.7447$: about a third of the neutrons leak out, which is why the boundary treatment matters as much as it does below.

Boundary treatment. Table 5 exercises the second branch of (27) against a second, independent analytic route — $u \cot u = 1 - u/(B\ell_0)$ from `critical_dimensions_applied_bc`, which solves the condition on the flux shape instead of extrapolating. The numerical Robin radius reproduces it to 1.5×10^{-6} relative, the same discretisation error as the extrapolated case, confirming the boundary coefficient in both of its limits.

The two treatments are not interchangeable, though: they agree to first order in $B\ell_0$ and separate as the system shrinks, from 0.26 % apart at $c = 1.02$ to 4.5 % at $c = 2$. That gap is not solver error but the genuine ambiguity in what "the surface" means for a system a few mean free paths across.

Table 5: Extrapolated zero at $R + z_0$ against the condition $\phi + z_0\phi' = 0$ applied at R, classical diffusion, in mean free paths.

c	extrapolated	Robin	Robin analytic	Robin − extrapolated
1.02	12.158813	12.126930	12.126949	−0.26 %
1.10	5.069062	5.007865	5.007874	−1.21 %
1.50	1.898429	1.821650	1.821653	−4.04 %
2.00	1.147130	1.095638	1.095640	−4.49 %

6 Question 5: Bare Critical Mass of U-235 and Pu-239

Question 5 is Question 4 with the one-group benchmark cross sections of Sood, Forster & Parsons (2003), the mass following as $M_c = \rho\frac{4}{3}\pi R_c^3$. Both fissile materials have $\Sigma_t = 0.32640$ cm^{-1}, a mean free path of 3.0637 cm, so both critical spheres are about two mean free paths in radius.

Table 6: Bare critical radius and mass from the $k = 1$ search on a 400-cell mesh. Numerical and analytic radii agree to four decimals in cm.

Material	Approximation	c	ρ [g/cm^3]	R_c [cm]	$\Sigma_t R_c$	M_c [kg]
Pu-239	classical	1.50	15.7	5.8163	1.8984	12.940
Pu-239	asymptotic	1.50	15.7	5.1890	1.6937	9.188
U-235	classical	1.30	19.0	8.1031	2.6449	42.345
U-235	asymptotic	1.30	19.0	7.4339	2.4264	32.696

Because the mass goes as the cube of the radius, the moderate difference in relaxation length is strongly amplified: the classical radius exceeds the asymptotic one by 12.1 % (Pu-239) and 9.0 % (U-235), which becomes mass ratios of 1.41 and 1.30. Classical diffusion thus over-predicts the

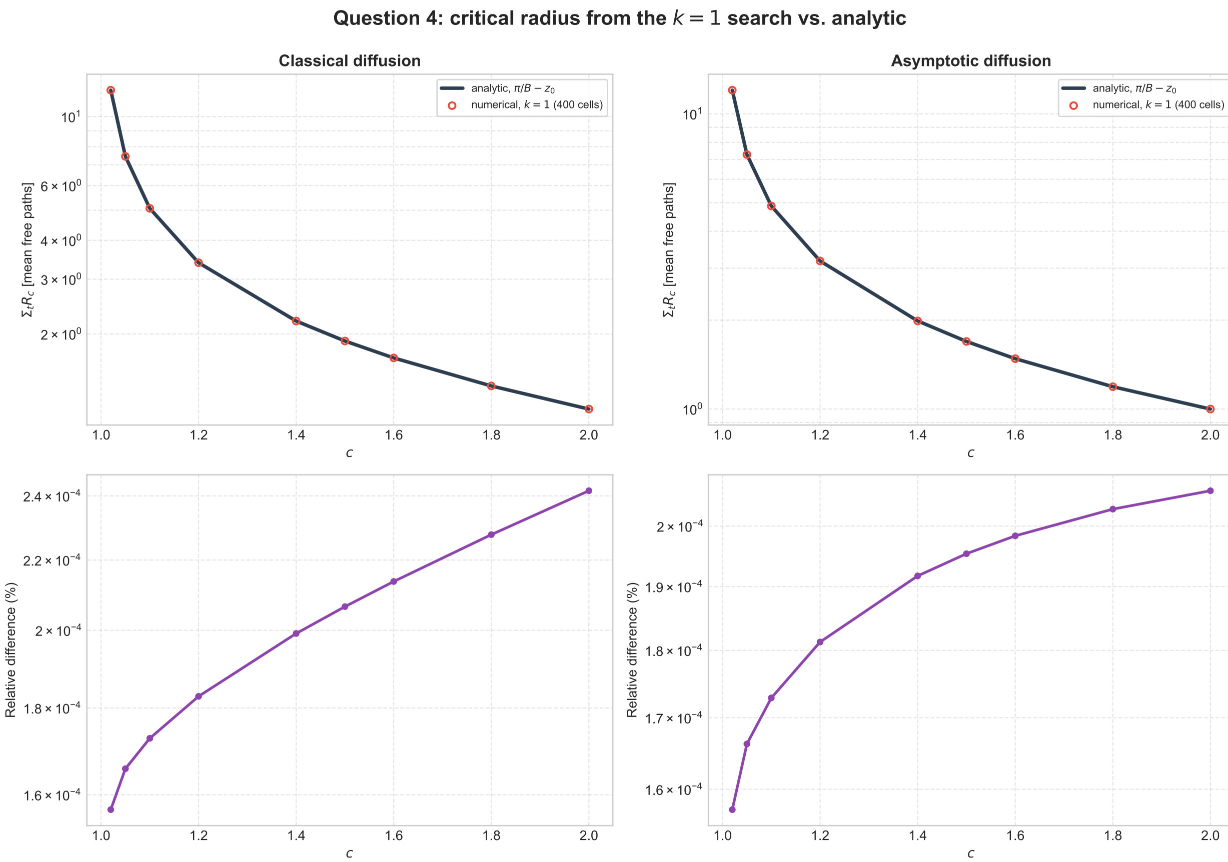

Figure 9: Critical radius from the $k = 1$ search (markers) against $\pi/B - z_0$ (solid), with the relative difference below.

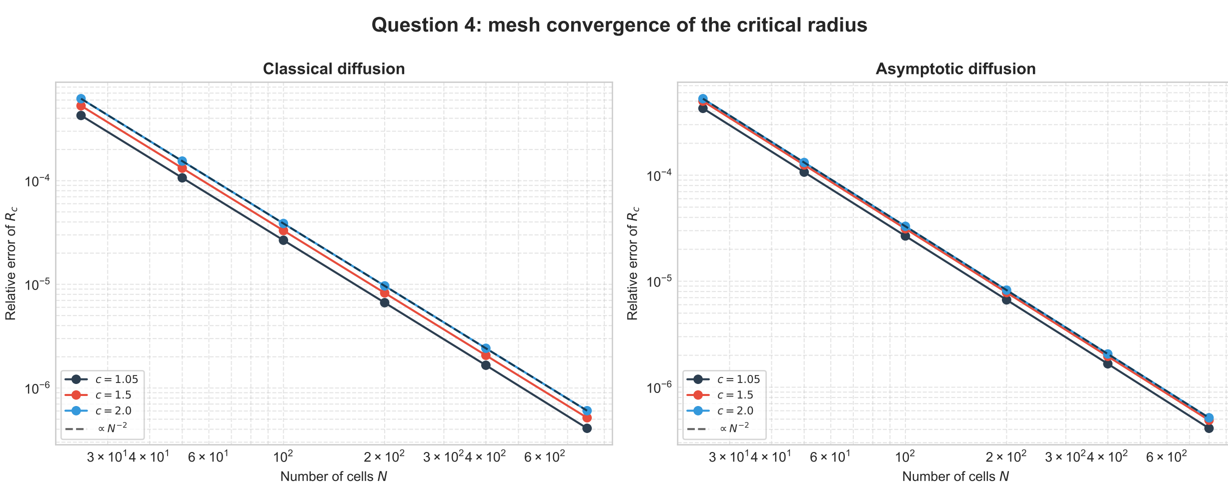

Figure 10: Mesh convergence of the critical radius, following the second-order guide $\propto N^{-2}$ at every refinement.

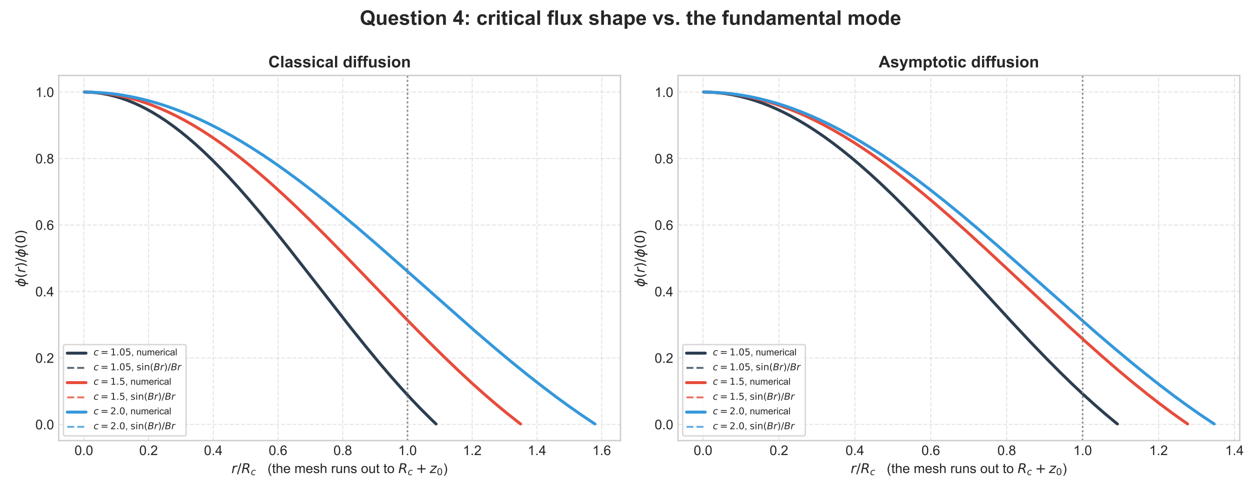

Figure 11: Converged critical flux (solid) against the fundamental mode $\sin(Br)/Br$ (dashed), normalised at the centre. The mesh continues past $r/R_c = 1$ to the extrapolated radius, where the flux is driven to zero.

critical mass by 30–40 % here, and worsens with c. The asymptotic values are the ones to quote, since asymptotic diffusion with the exact $z_0(c)$ reproduces the exact-transport relation.

6.1 Sensitivity to the Boundary Treatment

The masses above use the extrapolated zero. Applying the same condition directly at the physical surface instead moves them by far more than any numerical error, again because the mass goes as the cube of a radius only a couple of mean free paths long.

Table 7: Critical mass under the two boundary treatments. Both reproduce their own analytic result to 1.5×10^{-6}, so the spread is model uncertainty, not solver error.

Material	Approximation	M_c extrapolated [kg]	M_c Robin [kg]	difference
Pu-239	classical	12.940	11.432	−11.7 %
Pu-239	asymptotic	9.188	8.307	−9.6 %
U-235	classical	42.345	38.679	−8.7 %
U-235	asymptotic	32.696	30.212	−7.6 %

Both treatments reproduce their own analytic result, so this 8–12 % spread is model uncertainty rather than solver error, and it is comparable to the difference between the two diffusion approximations themselves. Any single quoted mass should be read with it in mind.

These are one-group, isotropic-scattering results, not predictions of real critical masses: at about two mean free paths the boundary layer is a large fraction of the system — roughly a third of the neutrons leak out — so diffusion of either flavour is well outside its regime, and the transport benchmark radii of Sood *et al.* are the appropriate reference. The remaining three benchmark rows (H_2O, Fe, Na) have $c \leq 1$ and are non-multiplying, so no bare critical sphere exists for them at any radius.

A note on the U-235 data. Two rows were in circulation. The assignment table ($\Sigma_c = 0.013056$, $\Sigma_s = 0.248064$, $\Sigma_t = 0.32640$) implies $c = 1.3000$, matching its quoted value, and is what is used above. The alternative row ($\Sigma_c = 0.015672$, $\Sigma_s = 0.180448$, $\Sigma_t = 0.26140$) is quoted with $c = 1.50$ but implies $c = 1.3646$, so it is not self-consistent; through the same code it gives $R_c = 8.9412$ cm, $M_c = 56.89$ kg classically and 8.1161 cm, 42.55 kg asymptotically — roughly 30 % more mass. Pu-239 is identical in both sources.

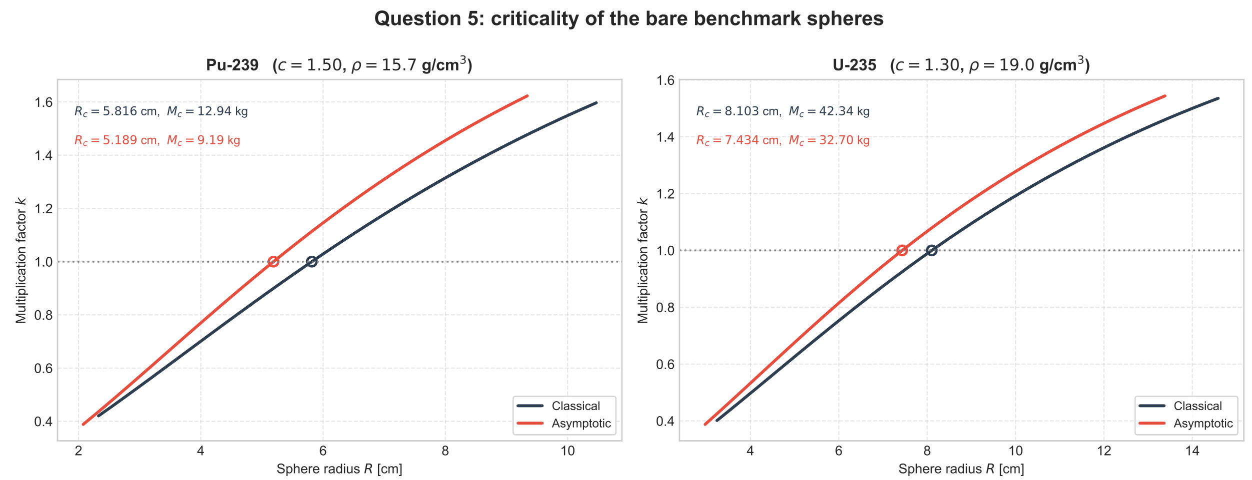

Figure 12: Multiplication factor against sphere radius for the two fissile materials, both approximations. The circles mark the $k = 1$ crossing.

A The Spherical Finite-Volume Scheme in Detail

An expansion of §5.1, following `src/homework1/spherical.py` line by line: what the mesh looks like, how `_banded_operator` turns it into a matrix, which index means what, and where the boundary conditions and the starting iterate enter.

A.1 The mesh and its indices

`_mesh` builds N uniform cells on $[0, R_{\text{out}}]$ with $h = R_{\text{out}}/N$. Two families of index run over it and they must not be confused:

- **Faces**, indexed $i = 0, \ldots, N$ — there are $N + 1$ of them, at radius $r_i = ih$, carrying areas $A_i = 4\pi r_i^2$ (`areas`, length $N + 1$).
- **Cells**, indexed $j = 0, \ldots, N - 1$ — there are N of them, cell j spanning $[r_j, r_{j+1}]$, with centre $\bar{r}_j = (j + \frac{1}{2})h$ (`centres`) and volume $V_j = \frac{4\pi}{3}(r_{j+1}^3 - r_j^3)$ (`volumes`), both of length N.

Cell j is bounded by face $i = j$ on the inside and face $i = j + 1$ on the outside, so the two numberings share their integers. The rule that removes every ambiguity below is: **a subscript on A, J or C is a face; a subscript on ϕ, V or S is a cell.** Where a relation ties a face to the two cells it separates, the same integer does double duty by construction — face i has cell $i - 1$ immediately inside it and cell i immediately outside. The unknown ϕ_j is the average flux over cell j; it lives at the centre, not at a face. In the half-index notation of §5.1, faces $i = j$ and $i = j + 1$ are what were written $j - \frac{1}{2}$ and $j + \frac{1}{2}$; the integer form is used here because it is what the code indexes.

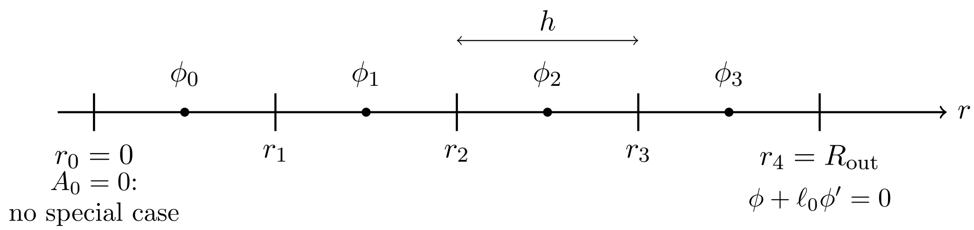

Figure 13: The cell-centred mesh for $N = 4$. Faces (ticks) carry areas, cells (dots) carry the unknowns. No unknown sits on a boundary, so both ends are face conditions rather than equations of their own.

A.2 From the cell balance to a matrix row

Rather than assume the balance (26), derive it from (24): multiply by the volume element $4\pi r^2\,dr$ and integrate across cell j, from face r_j to face r_{j+1}. The leading term is the one that matters, and the r^2 of the volume element is exactly what cancels the $1/r^2$ of the operator:

$$\begin{aligned}\int_{r_j}^{r_{j+1}} \left[-D\,\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right)\right]4\pi r^2\,dr &= -4\pi D\int_{r_j}^{r_{j+1}}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right)dr \\ &= -4\pi D\left[r^2\frac{d\phi}{dr}\right]_{r_j}^{r_{j+1}} \;=\; A_{j+1}J_{j+1}-A_jJ_j\end{aligned} \tag{31}$$

using $A = 4\pi r^2$ and $J = -D\,d\phi/dr$. The integral of a divergence has collapsed onto the two faces — the divergence theorem for a shell — which is why nothing is ever evaluated at $r = 0$, where the operator itself is singular.

The other two terms integrate to volume averages by definition of the unknown, $\phi_j \equiv V_j^{-1}\int_{\text{cell } j}\phi\,dV$:

$$\int_{r_j}^{r_{j+1}}\Sigma_a\phi\,4\pi r^2dr = \Sigma_a V_j\phi_j, \qquad \int_{r_j}^{r_{j+1}}\frac{\nu\Sigma_f\phi}{k}\,4\pi r^2dr = S_jV_j, \qquad S_j \equiv \frac{\nu\Sigma_f\phi_j}{k} \tag{32}$$

Collecting the three gives (26), with the half-indices written as the integer face labels of §A:

$$A_{j+1}J_{j+1} - A_jJ_j + \Sigma_aV_j\phi_j = S_jV_j \tag{33}$$

Nothing has been approximated yet: this is an exact statement about cell averages and true face currents, for any N. **The discretisation enters only in the next step**, where the face current is replaced by a difference between neighbouring cell centres — which is also where the second-order accuracy of §5.4 comes from. With Fick's law between adjacent centres, the current through face i — which separates cells $i-1$ and i, a distance h apart — is

$$J_i = -D\,\frac{\phi_i-\phi_{i-1}}{h} \qquad\Longrightarrow\qquad A_iJ_i = -C_i\,(\phi_i-\phi_{i-1}), \qquad \boxed{C_i \equiv \frac{DA_i}{h}} \tag{34}$$

The *conductance* C_i is a property of a face, not of a cell — this is the whole idea of the layout.

Getting from (33) to the matrix row takes three small steps, and the index change happens in the first of them. Cell j has two faces, so the face relation is evaluated twice: at the inner face $i = j$, whose neighbours are cells $j-1$ and j, and at the outer face $i = j+1$, whose neighbours are cells j and $j+1$. That substitution is where a face index becomes a pair of cell indices:

$$A_jJ_j = -C_j\,(\phi_j-\phi_{j-1}), \qquad A_{j+1}J_{j+1} = -C_{j+1}\,(\phi_{j+1}-\phi_j) \tag{35}$$

Putting both into (33), the inner face carrying the minus sign of the balance and so flipping its bracket,

$$\underbrace{-C_{j+1}\,(\phi_{j+1}-\phi_j)}_{\text{out through face } j+1} + \underbrace{C_j\,(\phi_j-\phi_{j-1})}_{-(\text{in through face } j)} + \Sigma_a V_j \phi_j = S_j V_j \tag{36}$$

Expanding the brackets and collecting the three unknowns ϕ_{j-1}, ϕ_j, ϕ_{j+1}:

$$-C_j\,\phi_{j-1} + (C_j + C_{j+1} + \Sigma_a V_j)\,\phi_j - C_{j+1}\,\phi_{j+1} = S_j V_j \tag{37}$$

So cell j couples to each neighbour through the conductance of the face it *shares* with that neighbour — C_j to the left, C_{j+1} to the right — and both of those same two conductances reappear on the diagonal, because every neutron leaving through a face also leaves the cell it came from. Each face appears in exactly the two rows it separates, with the same magnitude and opposite sign. That single fact gives the matrix all three of its useful properties: it is tridiagonal, it is symmetric, and it conserves neutrons exactly (§A.5).

A.3 The two ends, and why neither needs a branch

The origin. Face 0 sits at $r = 0$, so $A_0 = 0$ and therefore $C_0 = 0$. The ϕ_{-1} term in (37) vanishes on its own for cell $j = 0$, and no current crosses the centre. The symmetry condition $\phi'(0) = 0$ is thus imposed *exactly*, by geometry, with no code path of its own — the reason `_banded_operator` can index a plain array at $j = 0$.

The outer surface. Face N has no cell beyond it. The surface flux ϕ_s is eliminated between the boundary condition $\phi + \ell_0\phi' = 0$ and the one-sided difference over the half cell $h/2$ that separates $\bar{r}_{N-1}$ from the surface:

$$J_s = -D\phi'(R_{\text{out}}) = \frac{D\phi_s}{\ell_0}, \qquad J_s = -D\,\frac{\phi_s - \phi_{N-1}}{h/2} \tag{38}$$

Equating and solving,

$$\phi_s = \phi_{N-1}\,\frac{\ell_0}{\ell_0 + h/2} \qquad \Longrightarrow \qquad J_s = \frac{D\,\phi_{N-1}}{\ell_0 + h/2}, \qquad C_N = \frac{DA_N}{\ell_0 + h/2} \tag{39}$$

which is the single line assigning `conductance[-1]`, the code writing ℓ_0 as the variable `l0` (a lower-case L, not a one). Because ℓ_0 appears only in the denominator, $\ell_0 = 0$ is not a special case: it gives $\phi_s = 0$ and $C_N = DA_N/(h/2)$, the zero-flux limit. The two treatments of `_outer_boundary` are then just two choices of (R_{out}, ℓ_0): the extrapolated zero uses $(R + z_0,\, 0)$, and the Robin condition uses $(R,\, z_0)$.

Note where C_N ends up: on the *diagonal* of the last row only. It removes neutrons from cell $N-1$ without coupling it to anything, which is exactly what leakage out of the system is.

A.4 The banded storage, index by index

Both indices of the matrix are cells: M_{mn} couples cell m to cell n. `scipy.linalg.solve_banded((1, 1), ab, b)` expects it in diagonal-ordered form, meaning

$$\texttt{ab}[\,1 + m - n,\; n\,] = M_{mn}, \qquad |m - n| \le 1 \tag{40}$$

so row 0 of `ab` holds the superdiagonal, row 1 the diagonal, row 2 the subdiagonal — each stored in the *column* of the entry, which is why the superdiagonal starts one slot in and the subdiagonal stops one slot early. The three assignments read:

code	meaning
`ab[0, 1:]  = -conductance[1:-1]`	$M_{n-1,n} = -C_n, \quad n = 1..N-1$
`ab[1, :]  = conductance[:-1] + conductance[1:] + sigma_a * volumes`	$M_{jj} = C_j + C_{j+1} + \Sigma_a V_j$
`ab[2, :-1] = -conductance[1:-1]`	$M_{n+1,n} = -C_{n+1}, \quad n = 0..N-2$

The face index survives in the conductance: the entry coupling cells $n-1$ and n is $-C_n$, the conductance of face $i = n$, which is precisely the face they share. The slices are the whole trick, and each says something physical:

- `conductance[:-1]` is $C_0 \dots C_{N-1}$, the *inner* face of each cell; `conductance[1:]` is $C_1 \dots C_N$, the *outer* face of each cell. Element j of their sum is the pair of faces bounding cell j — the diagonal.
- `conductance[1:-1]` is $C_1 \dots C_{N-1}$, the $N-1$ *interior* faces, i.e. precisely those that separate two cells. It is dropped into both off-diagonals unchanged, which is why M comes out symmetric.
- The two excluded entries are the two boundaries: $C_0 = 0$ (nothing to couple across the origin) and C_N (leakage, coupling to nothing).

For $N = 4$, with $C_0 = 0$ already substituted,

$$M = \begin{pmatrix} C_1 + \Sigma_a V_0 & -C_1 & 0 & 0 \\ -C_1 & C_1 + C_2 + \Sigma_a V_1 & -C_2 & 0 \\ 0 & -C_2 & C_2 + C_3 + \Sigma_a V_2 & -C_3 \\ 0 & 0 & -C_3 & C_3 + C_4 + \Sigma_a V_3 \end{pmatrix} \tag{41}$$

is stored as

$$\texttt{ab} = \begin{pmatrix} * & -C_1 & -C_2 & -C_3 \\ C_1 + \Sigma_a V_0 & C_1 + C_2 + \Sigma_a V_1 & C_2 + C_3 + \Sigma_a V_2 & C_3 + C_4 + \Sigma_a V_3 \\ -C_1 & -C_2 & -C_3 & * \end{pmatrix} \tag{42}$$

the two $*$ entries being the unused corners, left as the zeros `np.zeros` put there.

A.5 Two properties to check the assembly against

Symmetry and definiteness. $M = M^\top$ because both off-diagonals are the same array, and the diagonal dominates strictly once $\Sigma_a > 0$ or $C_N > 0$. So M is positive definite, the LU factorisation inside `solve_banded` needs no pivoting, and the solve is $O(N)$.

Row sums are the neutron balance. Summing row j of (37), the conductances cancel in pairs and leave $\Sigma_a V_j$ for every interior row — and $\Sigma_a V_{N-1} + C_N$ for the last. In matrix form,

$$M\mathbf{1} = \Sigma_a \mathbf{V} + C_N \mathbf{e}_{N-1} \tag{43}$$

that is: applied to a flat flux, the operator returns absorption plus surface leakage and nothing else. This is the discrete identity behind the balance check of §5.4; it holds by construction rather than by luck, which is why that check probes the boundary coefficient (39) and little else.

A.6 The source, k, and what plays the role of an initial condition

Being a steady eigenvalue problem, this has no initial condition in the time-dependent sense. What it has instead is a *starting iterate* for the power iteration, and it is worth being clear about which parts of the setup are physics and which are merely a starting guess:

- **Baked into the matrix, once:** both boundary conditions. $C_0 = 0$ and C_N are assembled before the first sweep, so every iterate — converged or not — satisfies them exactly. They are never re-imposed inside the loop.
- **The right-hand side** is the *volume-integrated* fission source $F_j = \nu\Sigma_f \phi_j V_j$, divided by k; the factor V_j is what makes it dimensionally consistent with the C_i's, which are integrated currents. Compare (37): the source term is $S_j V_j$, not S_j.
- **The starting guess** is a flat flux $\phi_j^{(0)} = 1$ with $k^{(0)} = 1$. Power iteration converges to the fundamental mode from any start with a non-zero component along it, which a strictly positive vector certainly has, so the converged k and shape do not depend on this choice — only the number of sweeps does.
- **The renormalisation** $\phi \leftarrow \phi / \max \phi$ after each sweep is likewise not physics: the eigenvector is defined only up to scale, and both integrals in the k update are homogeneous in ϕ, so the factor cancels. It is there to keep the iterate at $O(1)$ so that the convergence test does not lose its significant digits.

The one quantity that does carry a physical scale through the whole calculation is the radius R: it fixes R_{out}, hence h, hence every area, volume and conductance. That is the sense in which the outer root search of `critical_radius` sits outside everything described here — each evaluation of $k(R)$ rebuilds the mesh and the matrix from scratch.